

FCU3

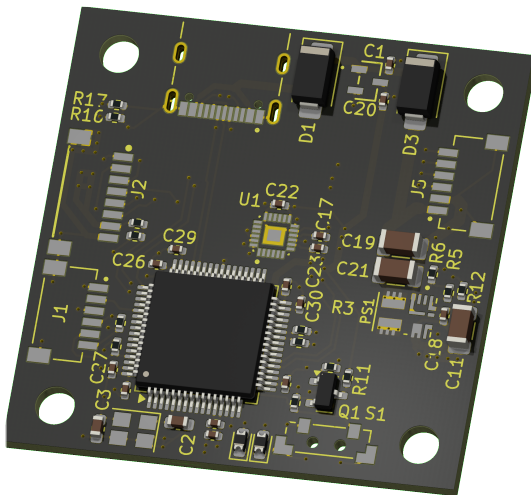
Variant: CHECKED

2025-12-13

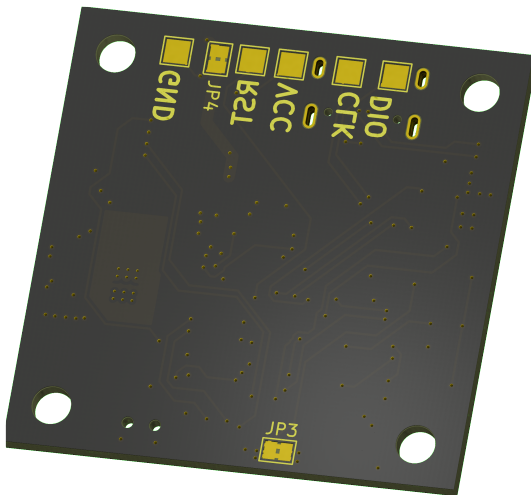
Rev + (Unreleased)

Page	Index	Page	Index	Page	Index	Page	Index
1	Cover Page	11		21		31	
2	Block Diagram	12		22		32	
3	Project Architecture	13		23		33	
4	Microcontroller	14		24		34	
5	Inertial Measurement Unit	15		25		35	
6	Power	16		26		36	
7	10–26V to 9V2A Step-Down Converter	17		27		37	
8	Revision History	18		28		38	
9		19		29		39	
10		20		30		40	

TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.
PRELIMINARY - Close to final schematic.
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
RELEASED - A board with this schematic has been sent to production.

Date: 13-Dec-2025

DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

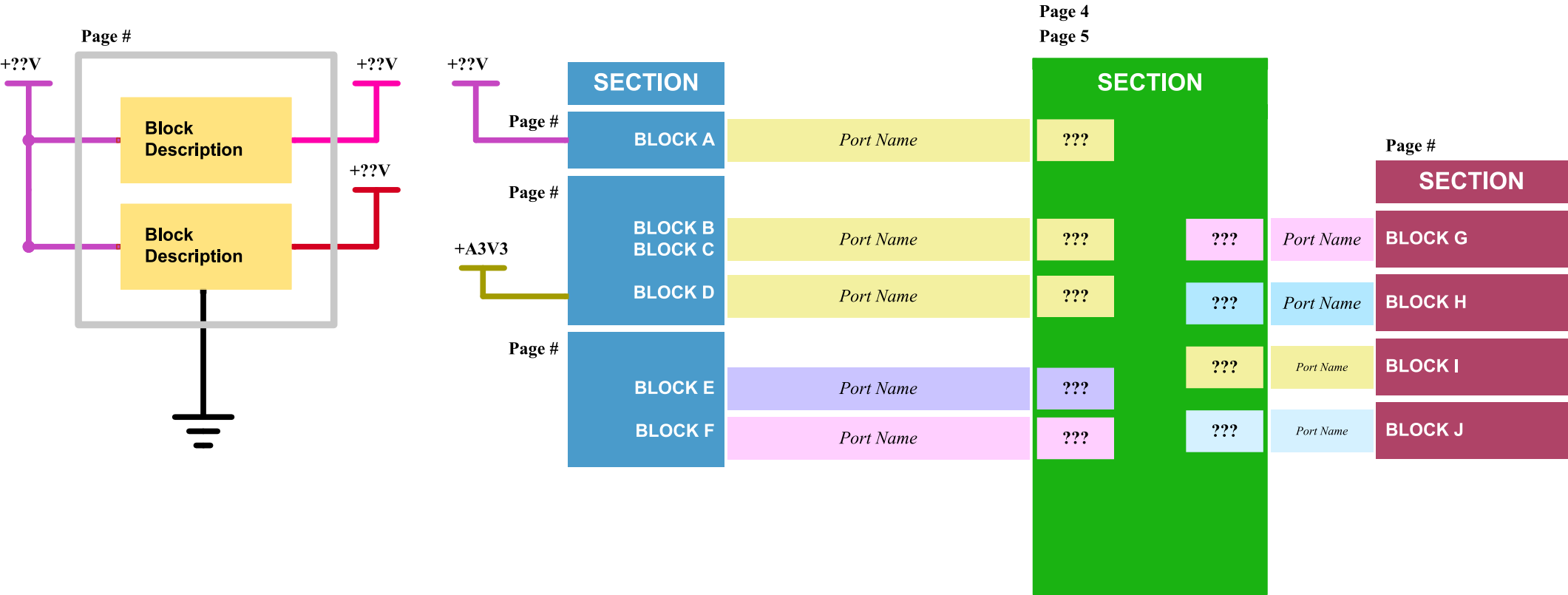
Example text for critical design notes.

LAYOUT NOTE:

Example text for critical layout guidelines.

	Comments:	Company:	Variant:	Git Hash:
		Kaldstrand Electronics	CHECKED	4dad50
		Board Name:	Project Name:	
	Sheet Title:	File Name:	Designer:	Date:
		FCU3.kicad_sch	Kaldstrand	2025-01-12
	Sheet Path:	Reviewer:	Size:	Revision:
	/		A3	+ (Unreleased)
				1 of 8

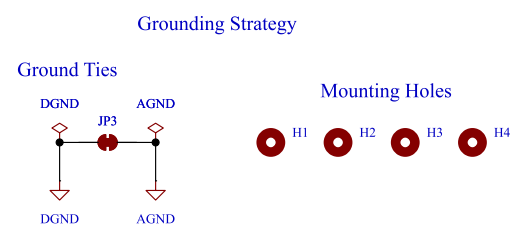
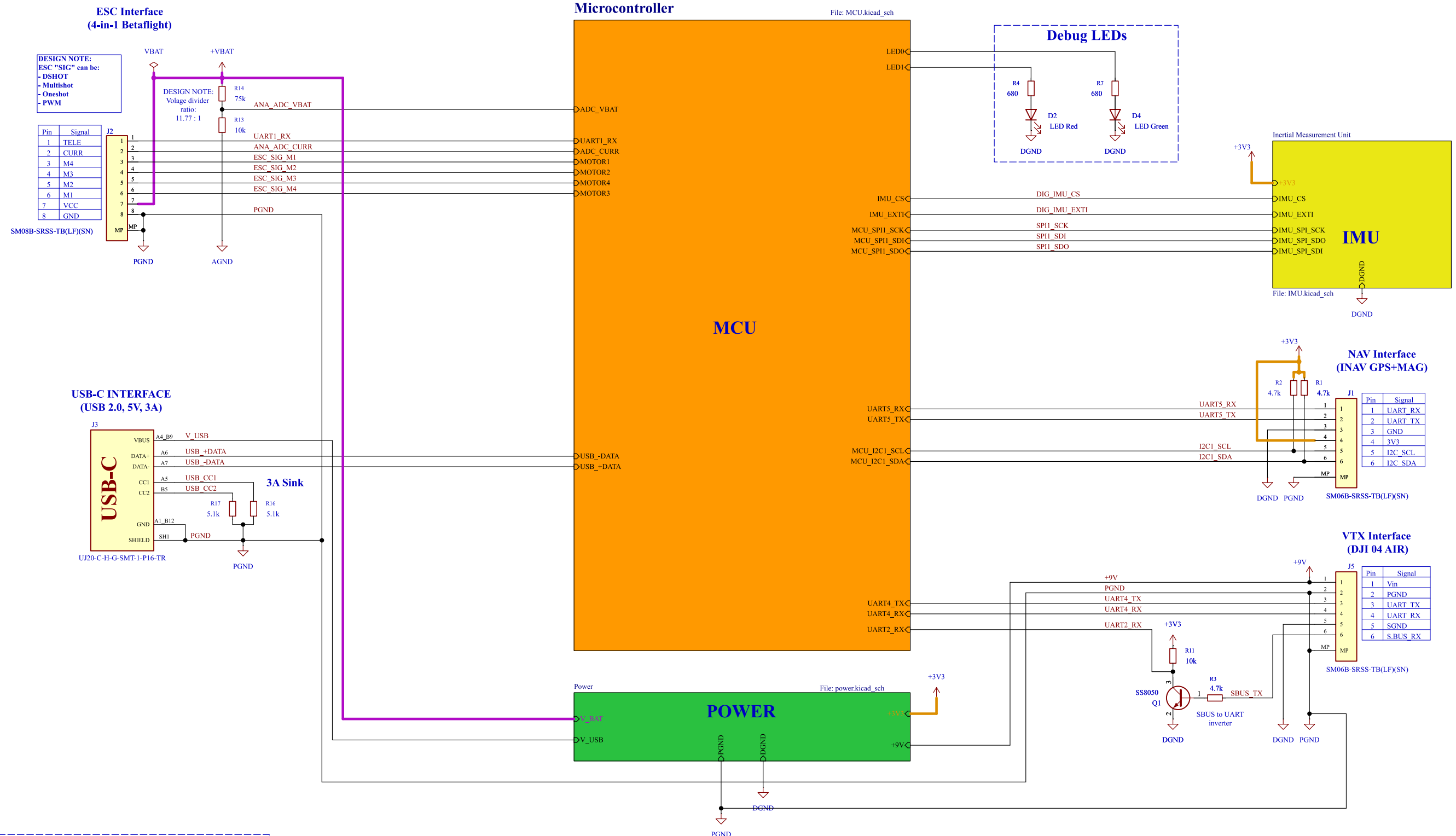
[2] Block Diagram



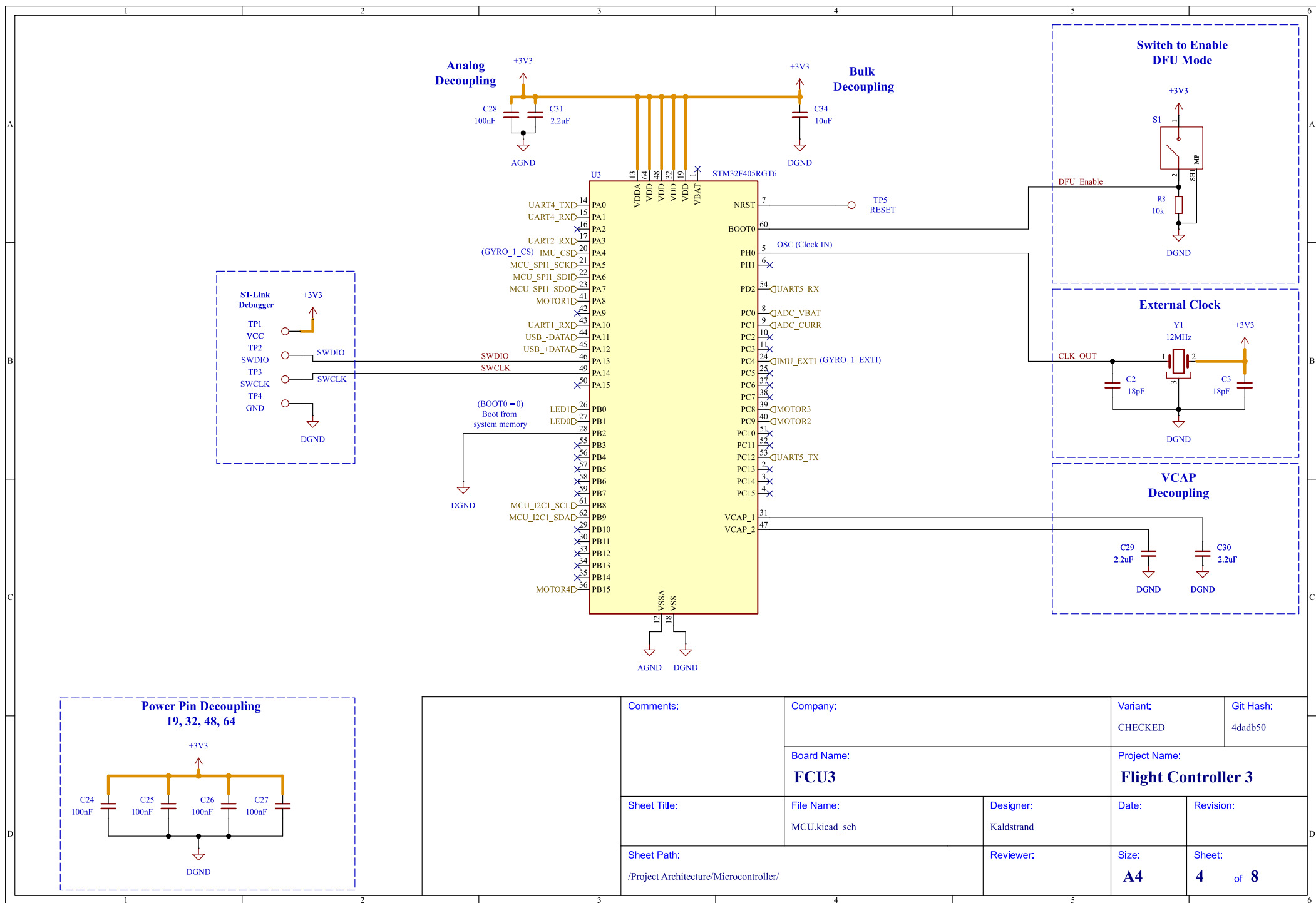
Target specifications:	
Input voltage:	?? - ?? V
Spec 2	??
Spec 3	??
Spec 4	??

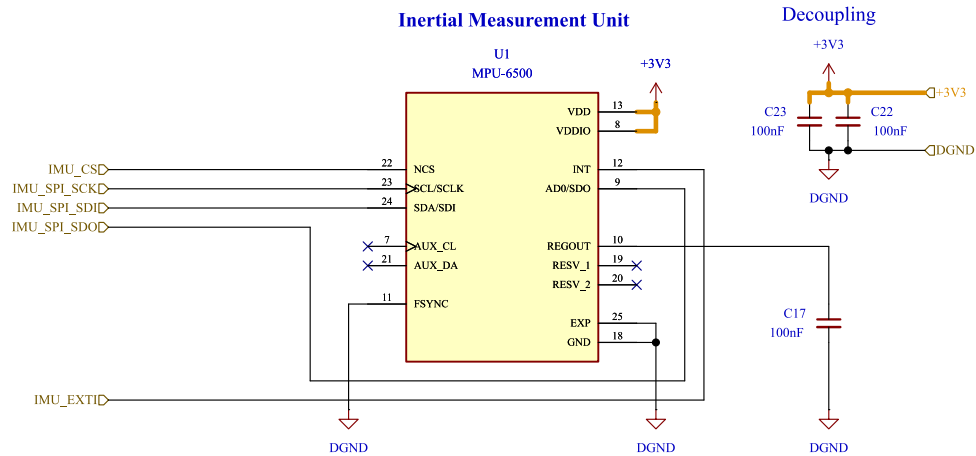
	Comments:	Company: Kaldstrand Electronics		Variant: CHECKED	Git Hash: 4dad50
		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Kaldstrand	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Block Diagram/			Reviewer:	Size: A3	Sheet: 2 of 8

[3] Project Architecture

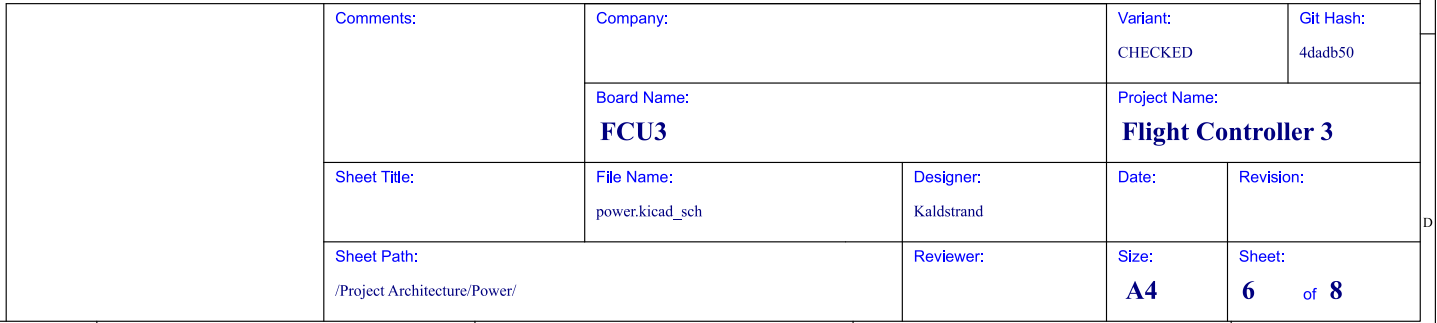


Comments:	Company: Kaldstrand Electronics		Variant: CHECKED	Git Hash: 4dad50
	Board Name: FCU3		Project Name: Flight Controller 3	
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Kaldstrand	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/		Reviewer:	Size: A3	Sheet: 3 of 8





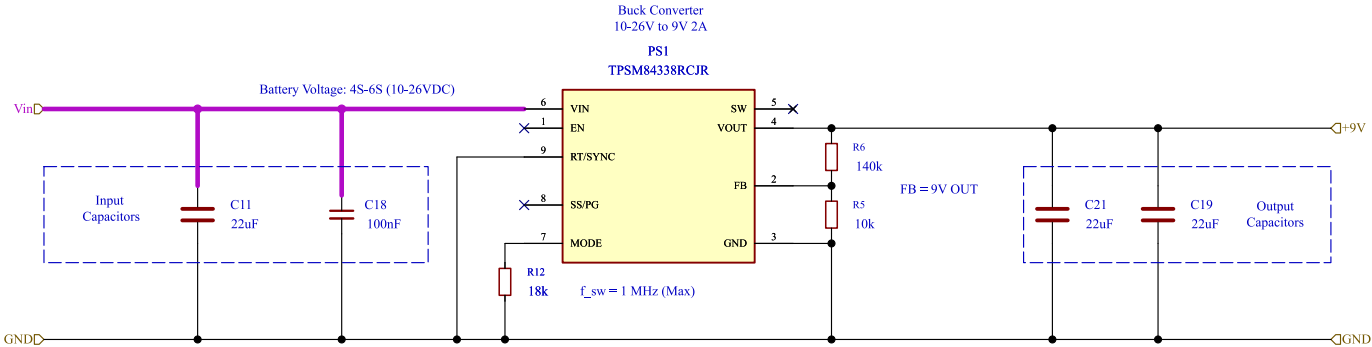
	Comments:	Company:	Variant: CHECKED	Git Hash: 4dad50
		Board Name: FCU3	Project Name: Flight Controller 3	
	Sheet Title:	File Name: IMU.kicad_sch	Designer: Kaldstrand	Date: Revision:
	Sheet Path: /Project Architecture/Inertial Measurement Unit/		Reviewer:	Size: A4 Sheet: 5 of 8



DESIGN NOTE:
 $R_{FBB} = (V_{OUT} - V_{REF}) / V_{REF} * R_{FBB}$
 $R_{FBB} = (9V - 0.6V) / 0.6V * 10k$
 $R_{FBB} = 140k$

 $V_{OUT} = 9V, V_{REF} = 0.6V, R11 = 10k, R6 = 140k$

To-Do:
Add design notes
explaining DC/DC settings.



	Comments:	Company:		Variant:	Git Hash:
				CHECKED	4dad50
		Board Name:		Project Name:	
		FCU3		Flight Controller 3	
	Sheet Title:	File Name:	Designer:	Date:	Revision:
		stepdown_10-26vin_9v2a.kicad_sch	Kaldstrand		
	Sheet Path:	Reviewer:	Size:	Sheet:	
	/Project Architecture/Power/10-26V to 9V2A Step-Down Converter/		A4	8 of 8	

[9] Revision History

	Comments:	Company: Kaldstrand Electronics		Variant: CHECKED	Git Hash: 4dad50
		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Kaldstrand	Date: 2025-01-12	Revision: + (Unreleased)
	Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 9 of 8